Dielectric Material

What is a Dielectric Material?

- A dielectric material is a type of insulator that becomes polarized when it comes in contact with an electric field.
- Although it is not a good conductor of electricity, it can easily support an electrostatic field.
- A dielectric material is an insulator which contains electric dipoles, that is where positive and negative charge are separated on an atomic or molecular level.
- When an electric field is applied, these dipoles align to the field, causing a net dipole moment that affects the material properties.
- Dielectric materials have two important functions;-
    1. They have the ability to increase the capacitance
    2. They have insulating ability.

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- They are found as solids, liquids and gases. A few examples of dielectric materials are:
- ***Dielectric Gas – Dry Air***
- ***Solid Dielectrics – Mica, Ceramic, Plastic and Glass***
- ***Dielectric Liquid – Distilled Water***
- The capacitance of a parallel plate capacitor with free space as an insulator is given by;

$$C = \frac{\varepsilon A}{d}, \qquad \varepsilon_r = {}^{\epsilon}/_{\epsilon_0}$$

- Capacitance is the ability to store charge across a potential difference.

Polarization in dielectrics

- When dielectrics are placed in an electric field, practically no current flows in them
- Because, unlike metals, they have no loosely bound, or free, electrons that may drift through the material.
- Instead, electric polarization occurs.
- The positive charges within the dielectric are displaced minutely in the direction of the electric field,
- And the negative charges are displaced minutely in the direction opposite to the electric field.
- This slight separation of charges, molecules, ions and dipoles, is called dielectric polarization.
- Reduces the electric field within the dielectric.

Cont…

- The presence of dielectric material affects other electrical phenomena.
- The force between two electric charges in a dielectric medium is less than it would be in a vacuum,
- While the quantity of energy stored in an electric field per unit volume of a dielectric medium is greater.
- The capacitance of a capacitor filled with a dielectric is greater than it would be in a vacuum.
- The effects of the dielectric on electrical phenomena are described on a large, or macroscopic, scale by employing such concepts as dielectric constant, permittivity, and electric polarization.

- Insulators are materials with very poor conductivity.
- Insulators are not very good at conducting either heat or electricity owing to the absence of loosely bound or freely moving charges in the atoms of the insulator.
- When insulators are placed in an electric field, practically no current flows in them, unlike in metals. Instead in some insulators, electric polarization occurs.
- A dielectric is an insulator that undergoes electric polarization on the application of the electric field.
- The charges in dielectric material do not move but only shifts slightly from the equilibrium position resulting in the dielectric polarization.

Various polarization processes:

When the specimen is placed inside a d.c. electric field, polarization is due to four types of processes….

1.Electronic polarization

2.Ionic polarization

3.Orientation polarization

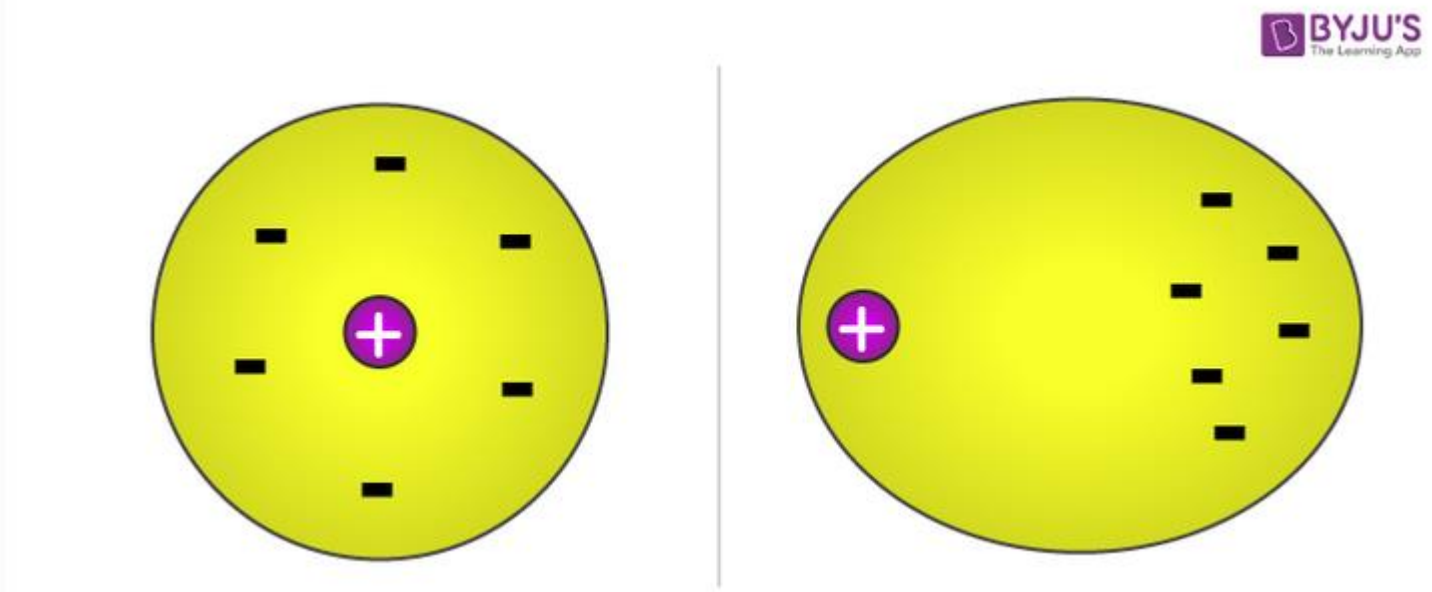
4.Space charge polarization

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- ***Polar and Non-Polar Material: Dielectric Material and Dipole Moment***, we examined the nature of bonds that decide the polarity of the molecules.
- Electronegativity and the structure of the atoms in the molecules decide whether it is a polar or nonpolar molecule.
- A nonpolar molecule refers to a molecule without a dipole.
- The charges in a nonpolar molecule are equally distributed.
- In spite of the lack of a dipole, a dielectric nonpolar material introduced in an electric field will be affected.
- In an electric field, the positive and the negative charges in a nonpolar molecule experience forces in opposite directions as a result of their opposite polarities.

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- This force causes the electron cloud of a nonpolar molecule to be displaced in the direction of the attraction.
- This displacement goes on until the attraction by the electric field is balanced by the internal forces of the molecule.
- Thus, in the presence on an electric field, even a nonpolar molecule experiences induced dipole moment.



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- This dipole moment is induced in the direction of the field and is directly proportional to the strength of the electric field the nonpolar material is subject to.
- Both polar and nonpolar molecule experience polarization on exposure to the electric field
- But the difference between a nonpolar and a polar molecule is that nonpolar molecules are induced with a dipole by current whereas polar molecules have permanent dipoles.
- Due to the induced nature of polarity, on the removal of the electric field, a nonpolar material loses its polarity are returned to its original state.

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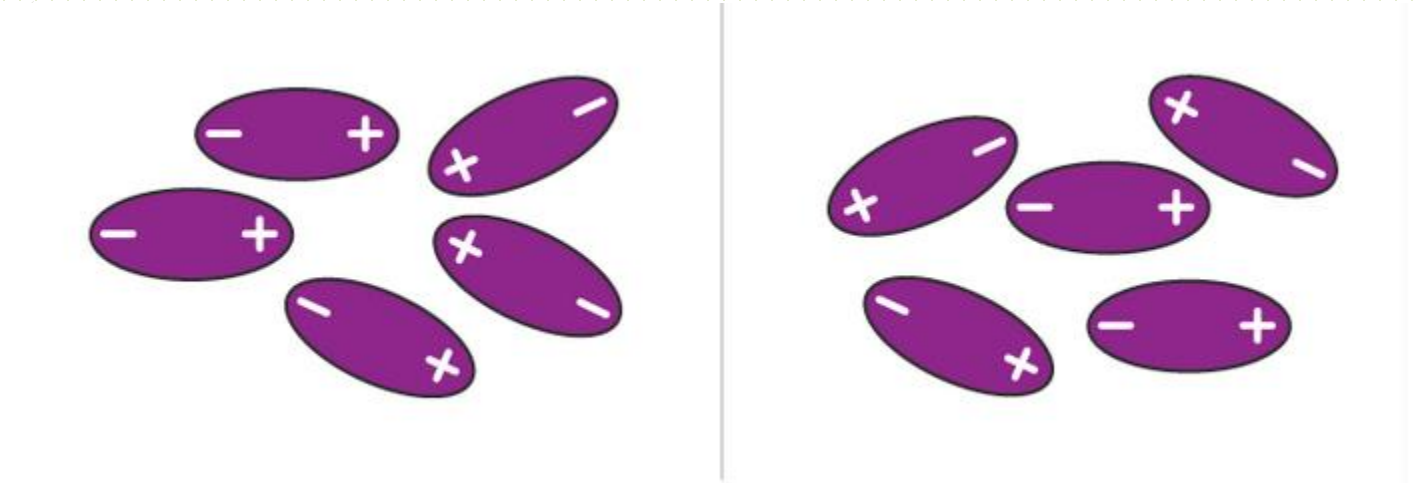
- Polar molecules undergo dipolar polarization which is also referred to as Orientation Polarization.
- A polar molecule, on the other hand, is already blessed with electric dipoles and this dipole is not induced.
- This dipole exists due to the bonds and the structure of a polar molecule. But we cannot utilize this already existing dipole moment right away.
- Due to thermal agitation, the dipoles in a polar material are oriented randomly.
- Therefore the dipole moment of the molecules in the material cancels out resulting in a net dipole moment of zero.
- We need to apply an electric field here as well, albeit for different purposes.

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- When an electric field is applied, the individual dipole moments align themselves in the direction of the electric field.
- This means that the bonds, their nature and their orientation remain constant and the polar molecule only rotates about its axis minutely to align itself.
- This alignment when summed up over all the molecules leads to a net dipole moment in the direction of the electric field.
- The extent to which the polar molecules get polarizes and align themselves is related to two factors;

1. the strength of the external field and
2. the thermal energy that breaks this alignment.

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Calculating Dipolar Polarization

- The already existing dipoles rotate to align with the electric field. It is also known as Orientation Polarization.
- Thus, irrespective of whether a material is polar or nonpolar, the application of electric field results in the creation of a net dipole moment across the material.
- The dipole moment per net unit volume is called Polarization For an ideal dielectric material,

Dielectric Constant

Dielectric Constant is the ratio between the permittivity of the medium to the permittivity of free space.

$$\varepsilon_r = \frac{\varepsilon}{\varepsilon_0}$$

$\epsilon_0 = 8.85 \times 10^{-12}\ Fm^{-1}$

The characteristics of a dielectric material are determined by the dielectric constant and it has no units.

Electric Polarization

The process of producing electric dipoles by an electric field is called polarization in dielectrics.

Polarizability:

The induced dipole moment per unit electric field is called Polarizability.

The induced dipole moment is proportional to the intensity of the electric field.

$$\mu \propto E$$

$$\mu = \alpha E$$

$$\alpha \rightarrow \text{polarizabi lity constant}$$

Is a Polarizability constant

Electric flux Density (D):

Electric flux density is defined as charge per unit area and it has same units of dielectric polarization.

Electric flux density D at a point in a free space or air in terms of Electric field strength is

$$D_0 = \varepsilon_0 E \qquad -->(1)$$

At the same point in a medium is given by

$$D = \varepsilon E \qquad -->(2)$$

As the polarization measures the additional flux density arising from the presence of material as compared to free space

$$\text{i.e,} \quad D = \varepsilon_0 E + P \qquad -->(3)$$

Using equations 2 & 3 we get

$$\varepsilon \mathrm{E} = \varepsilon_0 \mathrm{E} + \mathrm{P}$$

$$(\varepsilon - \varepsilon_0)\,\mathrm{E} = \mathrm{P}$$

$$\text{(or)} \quad (\varepsilon_r . \varepsilon_0 - \varepsilon_0)\,\mathrm{E} = \mathrm{P}$$

$$(\varepsilon_r - 1)\varepsilon_0 . E = \mathrm{P}$$

Electric susceptibility:

The polarization vector P is proportional to the total electric flux density and direction of electric field.

Therefore the polarization vector can be written

$$\overline{P} = \varepsilon_0 \chi_e E$$

$$\chi_e = \frac{\overline{P}}{\varepsilon_0 E}$$

$$= \frac{\varepsilon_0 (\varepsilon_r - 1) E}{\varepsilon_0 E}$$

$$\chi_e = \varepsilon_r - 1$$

Applications of Dielectric Materials

- The dielectric material is largely used in the manufacturing of capacitors.
- Dielectric due to their unique capacity for electrical polarization.
- We have already discussed the phenomenon of the induced dipole in nonpolar materials and the alignment of the dipole in the polar material.
- The dielectric sample contains a very large number of dipoles distributed throughout its body.
- When subjected to an electric field, the positive end of one dipole sits next to the negative of the neighboring dipole, the positive end of which sits next to the negative terminal of the other dipole and so on.

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- They form a chain of alternating positive and negative polarities throughout the material.
- This also the case with nonpolar materials and this alignment leads to the creation of an electric field.
- This electric field which is set up due to the net dipole moment of the material opposes the external electric field which reduces the electric field built up by a capacitor during charging.
- This means that we can now accumulate more charges to bring back the electric field to its original intensity.
- This means that the capacity of a capacitor is effectively increased by introducing a dielectric medium between the plates of the capacitor.

Lorentz field

- **Lorentz force**, the force exerted on a charged particle *q* moving with velocity ***v*** through an electric field ***E*** and magnetic field ***B***.
- The entire electromagnetic force ***F*** on the charged particle is called the Lorentz force (after the Dutch physicist Hendrik A. Lorentz) and is given by:

$$\boldsymbol{F} = q\boldsymbol{E} + q\boldsymbol{v} \times \boldsymbol{B}.$$

- The first term is contributed by the electric field. The second term is the magnetic force and has a direction perpendicular to both the velocity and the magnetic field.
- An interesting result of the Lorentz force is the motion of a charged particle in a uniform magnetic field.

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- If ***v*** is perpendicular to ***B*** (i.e., with the angle φ between ***v*** and ***B*** of 90°), the particle will follow a circular trajectory with a radius of

$$r = \frac{mv}{qB}$$

- If the angle φ is less than 90°, the particle orbit will be a helix with an axis parallel to the field lines.
- If φ is zero, there will be no magnetic force on the particle, which will continue to move undeflected along the field lines.
- Charged particle accelerators like cyclotrons make use of the fact that particles move in a circular orbit when ***v*** and ***B*** are at right angles.

- For each revolution, a carefully timed electric field gives the particles additional kinetic energy, which makes them travel in increasingly larger orbits.
- When the particles have acquired the desired energy, they are extracted and used in a number of different ways, from studies of subatomic particles to the medical treatment of cancer.